Variables and Constants in Algebraic Expressions

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Algebraic Expression

An **algebraic expression** is a mathematical phrase made up of numbers, variables, and operations such as addition, subtraction, multiplication, and division.

$$3x + 8$$

Variable

A **variable** is a letter or symbol that represents a number whose value may change or is not yet known. Common variables include

$$x,\ y,\ a,\ b,\ n$$

Constant

A **constant** is a number whose value remains fixed in a given expression. Examples include

$$2,\ 5,\ -7,\ 12$$

Memory Trick

Variables vary.

Constants stay **constant**.

Letter $\rightarrow$ variable

Number $\rightarrow$ constant

Coefficient

The number multiplied by a variable is called its **coefficient**.

$$3x + 8$$

The number 3 is the coefficient of x.

Constant Term

A **constant term** contains no variable.

$$4x + 11$$

The constant term is 11.

Parts of an Expression

$$\underbrace{3}_{\text{coefficient}}\ \underbrace{x}_{\text{variable}} + \underbrace{8}_{\text{constant}}$$

Examples

Example 1

Identify the variable and constants.

$$3x + 8$$

Variable: x

Constants: 3 and 8

The number 3 is the coefficient of x.

Example 2

Identify the variable and constants.

$$5y - 4$$

Variable: y

Constants: 5 and 4

The minus sign indicates subtraction.

Example 3

Identify the variables and constants.

$$7a + 2b + 9$$

Variables: a and b

Constants: 7, 2, and 9

The numbers 7 and 2 are coefficients of the variables.

Example 4

Find the constant term.

$$4x + 11$$

The constant term is 11 because it has no variable.

Summary Table

Expression	Variable(s)	Constants
$3x + 8$	x	$3,\ 8$
$5y - 4$	y	$5,\ 4$
$7a + 2b + 9$	$a,\ b$	$7,\ 2,\ 9$

Practice Exercises

Practice 1

Identify the variable(s) and constant(s).

$$5x + 12$$

Practice 2

Identify the variable(s) and constant(s).

$$8 - y$$

Practice 3

Identify the variable(s) and constant(s).

$$4a + 3b + 7$$

Practice 4

Identify the variable(s) and constant(s).

$$15m - 6$$

Practice 5

Identify the variable(s) and constant(s).

$$2p + 9q + 11$$

Practice 6

Identify the variable(s) and constant(s).

$$10r - 3s + 5$$

Practice 7

Identify the variable(s) and constant(s).

$$7x + 2x + 9$$

Practice 8

Identify the variable(s) and constant(s).

$$6a - 4b + 13$$

Practice 9

Identify the variable(s) and constant(s).

$$20 - 5n$$

Practice 10

Identify the variable(s) and constant(s).

$$3u + 8v - 2w + 14$$

Activities

Activity 1

Identify the variable(s) and constant(s).

$$7x + 15$$

Activity 2

Identify the variable(s) and constant(s).

$$11 - y$$

Activity 3

Identify the variable(s) and constant(s).

$$5a + 6b + 9$$

Activity 4

Identify the variable(s) and constant(s).

$$18m - 7$$

Activity 5

Identify the variable(s) and constant(s).

$$4p + 12q + 3$$

Activity 6

Identify the variable(s) and constant(s).

$$9r - 2s + 8$$

Activity 7

Identify the variable(s) and constant(s).

$$5x + 3x + 14$$

Activity 8

Identify the variable(s) and constant(s).

$$8a - 6b + 17$$

Activity 9

Identify the variable(s) and constant(s).

$$25 - 7n$$

Activity 10

Identify the variable(s) and constant(s).

$$6u + 4v - 3w + 19$$

Thank you!